

MathKix

Question Review Guide

This document is for freelance reviewers working in the MathKix admin dashboard. It covers the full decision framework, curriculum cheat sheet, question type rules, flag definitions, and edge cases you will encounter.

Your task in one sentence

For each question: read it, check it against this guide, then click **Approve** or **Flag**.

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1. How the Dashboard Works

Log in at mathkix.com/admin → click **Questions** in the left sidebar. You will see a card for each question. The default filter shows **AI Flagged** questions first — these were pre-screened by an AI and need your human judgement to confirm or override.

Filters

AI Flagged — AI found a potential issue. Review these first.

Unreviewed — Not yet seen by anyone. Work through these after AI Flagged.

Approved — Already approved. You can revisit if needed.

Flagged — Human-confirmed issues. Skip unless re-checking.

Actions

Approve (→ or L key) — Question is correct, clear, and appropriate. Move on.

Flag (← or H key) — Opens a form. Fill in:

- **What's wrong?** (required) — Brief description of the issue.
- **Suggested fix** (optional but very helpful) — What the question should say or what the answer should be.

Tip — keyboard shortcuts

Use → to approve and ← to flag. Use ↑ / ↓ to navigate without deciding. This makes reviewing fast once you are in a rhythm.

2. Decision Framework

Go through these checks in order. Stop at the first failure and flag.

1	Is the correct answer mathematically right? Work it out yourself. If it is wrong, flag it as Wrong Answer with the correct value in Suggested Fix.
2	Can the student actually enter this answer using the UI type? See Section 3 for the rules per type. This is the most common mechanical failure.
3	Is the question clear and unambiguous? Read it as a child of that grade. If it could mean two different things, flag as Ambiguous Wording .
4	Does the question have enough information to be solved? If it references a picture/diagram/chart but no visual is shown, flag as Missing Visual . If a number or unit is missing, flag as Unanswerable .
5	Does the difficulty match the grade level standard? (Check the cheat sheet in Section 6) Always check the difficulty stars before flagging grade mismatch. Hard (★★★) questions at a grade level are expected to push the upper boundary of that standard.
6	For multiple choice: are the wrong options (distractors) reasonable? Distractors should reflect common mistakes — not be obviously silly or identical to each other. Flag as Weak Distractors if a child could eliminate wrong answers without knowing the math.
7	Is the language appropriate for the grade? Grade 1-2: vocabulary should be simple, sentences short. Grade 3-5: can be more complex. Flag as Ambiguous Wording if language seems wildly off.
When in doubt — approve If you have spent more than 60 seconds on a question and are still unsure, approve it. The goal is to catch clear errors, not to debate edge cases. Flag only when you are confident something is wrong.	

3. Question Types & UI Rules

Each question has a type. The type determines what UI component the student uses to answer. If the correct answer cannot be physically entered using that UI, the question is broken — flag it as **UI Mismatch**.

Type	How the student answers	Valid correct_answer format	Common failures
multiple_choice	Taps one of 2–4 labelled buttons (A/B/C/D)	Must exactly match one of the option values (case-sensitive)	Values are 'twelve', '10', '14', '16' — fewer than 2 options
numeric	Types on a virtual number pad (digits 0–9 and a decimal point)	NO fractions, NO words	Answer is '1/2' — must use fraction type Answer is 'A half' — not a number Answer is 'seven' — not a number
fraction	Two separate boxes — numerator (top) and denominator (bottom)	Examples: '1/2', '3/4', '7/8' NO decimals, NO mixed numbers	Answer is '0.5' — must be '1/2' Answer is '1 1/2' — mixed numbers not supported Denominator is 0 (e.g. '3/0')

Multiple choice — extra checks

- The **correct_answer** field must exactly match one option **value** (not label). Labels are A/B/C/D. Values are what the student actually answers, e.g. '12'.
- There must be at least 2 options, ideally 4.
- No two options should have the same value.
- The correct option should not be obviously different in style or length from wrong options.

Fraction type — grade rule

- Fractions (NF domain) are a **Grade 3+ topic** in Common Core.
- A question tagged Grade 1 or Grade 2 with type 'fraction' is almost certainly a data error — flag as **Grade Mismatch + UI Mismatch**.
- Exception: Grade 2.G.A.3 asks students to partition shapes into equal shares (halves, thirds, fourths) — but the answer is described in words, not entered as a fraction. If a G2 question asks for a fraction input, flag it.

4. AI Flags — What They Mean & When to Trust Them

The AI pre-screened all questions and left flags where it suspected problems. It is helpful but not perfect.

Your human judgement always overrides the AI.

Flag	What it means	Trust?	What to do
UI Mismatch	Answer format doesn't match the UI type perfectly (e.g. fraction bars, decimal points, etc.)	High	Check Section Grids display. If confirmed, flag. These are usually clear-cut data errors.
Format Error	Multiple choice has fewer than 2 options, or options are confusing.	High	Confirm by looking at the options grid. If empty or only 1 option, flag.
Wrong Answer	AI thinks the stated correct answer is mathematically incorrect.	Medium	Work out the answer yourself. AI makes arithmetic mistakes. Approve if you agree.
Missing Visual	Question text says 'look at the diagram' or 'shown below' but no visual is attached.	High	Check if the question is self-contained without a visual. If it references something not in the question, flag.
Grade Mismatch	AI thinks the question is too hard or too easy for the grade.	Low	Always check the difficulty stars first. Hard (3-star) questions at a grade are expected.
Ambiguous Wording	Question could be interpreted multiple ways.	Medium	Read it as a child. If you can only see one interpretation, approve.
Weak Distractors	Wrong options are too obvious or nonsensical for multiple choice.	Low	Check if wrong options reflect plausible mistakes. If they do, approve — AI is often wrong here.
Unanswerable	Missing information needed to solve the problem.	Medium	Try to solve it yourself. If you can, approve.

Key principle on Grade Mismatch flags

The AI does not know the difficulty setting. A 3-star (hard) question labelled Grade 2 is EXPECTED to be at the upper edge of Grade 2 content — that is its purpose. Always check the stars before accepting a grade mismatch flag. The example '345 + 478' for Grade 2 hard is fine — it's 2.NBT.7.

5. Difficulty Levels

Every question has a difficulty from 1 (easy) to 3 (hard). This affects what content is acceptable.

Stars	Label	What to expect
★■	Easy	Straightforward application of the standard. Single step. Small numbers. No tricks. A student who just learned
★★■	Medium	Requires applying the standard in a slightly less obvious way. May have two steps or slightly larger numbers.
★★★	Hard	Pushes the upper boundary of the standard. Multi-step, larger numbers, word problems, or application in a new

6. Curriculum Cheat Sheet

Use this section to verify whether a question's content matches its standard code. Standard codes appear on the question card (e.g. '2.NBT.7'). The format is: **Grade.Domain.Standard**.

Domains: OA = Operations & Algebra | NBT = Number & Base Ten | NF = Fractions (Grade 3+) | MD = Measurement & Data | G = Geometry

Grade 1

Code	What the student should be able to do	Example question
1.OA.A.1	Add/subtract within 20 to solve word problems using objects, drawings, and equations	There are 8 apples. 3 are eaten. How many are left?
1.OA.A.2	Solve word problems adding three whole numbers whose sum is at most 20	Sarah has 2 red, 4 blue, and 5 green marbles. How many total?
1.OA.B.3	Apply commutative and associative properties of addition	If $4 + 6 = 10$, what does $6 + 4$ equal?
1.OA.B.4	Understand subtraction as an unknown-addend problem	$8 - 3 = \underline{\quad}$. What number plus 3 equals 8?
1.OA.C.6	Add and subtract within 20 fluently	What is $7 + 8$?
1.OA.D.7	Understand the equal sign (=) means both sides are the same	Is $5 + 3 = 4 + 4$ true or false?
1.NBT.B.2	Understand two-digit numbers as tens and ones	What does the digit 3 mean in the number 35?
1.NBT.B.3	Compare two-digit numbers using $>$, $=$, $<$	Which is greater: 47 or 74?
1.NBT.C.4	Add within 100 (two-digit + one-digit, two-digit + multiple of 10)	What is $34 + 20$?
1.NBT.C.5	Mentally find 10 more or 10 less than a two-digit number	What is 10 more than 56?
1.MD.B.3	Tell and write time in hours and half-hours	The clock shows 3:30. What time is it?
1.MD.C.4	Organize and interpret data with up to three categories	The chart shows 4 cats, 7 dogs, 2 fish. How many more dogs than cats?
1.G.A.1	Distinguish defining attributes of shapes (sides, angles)	Which shape has 3 sides?
1.G.A.3	Partition circles and rectangles into two and four equal shares	A circle is cut into 2 equal pieces. What is each piece called?

Grade 2

Code	What the student should be able to do	Example question
2.OA.A.1	Add/subtract within 100 to solve one- and two-step word problems	Lena has 45 stickers. She gives away 18. How many does she have left?
2.OA.B.2	Fluently add and subtract within 20 from memory	What is $13 - 7$?
2.OA.C.3	Determine if a group of objects has an odd or even number	Is 14 odd or even?
2.OA.C.4	Use addition to find total objects in rectangular arrays	A 3-row array has 4 items per row. How many total?
2.NBT.A.1	Understand 3-digit numbers as hundreds, tens, and ones	What does the digit 4 represent in 347?

Code	What the student should be able to do	Example question
2.NBT.A.3	Read and write numbers to 1,000 in multiple forms	Write 'five hundred sixty-two' in standard form.
2.NBT.A.4	Compare three-digit numbers using $>$, $=$, $<$	Which is less: 472 or 427?
2.NBT.B.5	Fluently add and subtract within 100	What is $67 - 29$?
2.NBT.B.7	Add and subtract within 1,000 using models and strategies	A school has 345 boys and 478 girls. How many students in all?
2.NBT.B.8	Mentally add or subtract 10 or 100 to/from any number	What is 100 more than 637?
2.MD.C.7	Tell and write time to the nearest 5 minutes	The clock shows 2:45. What time is it?
2.MD.C.8	Solve word problems with dollar bills, coins	If you have 3 quarters and 2 dimes, how many cents do you have?
2.G.A.1	Recognize and draw shapes with specific attributes	Which shape has 4 equal sides?
2.G.A.3	Partition rectangles/circles into halves, thirds, fourths	A pizza is cut into 4 equal slices. What fraction is one slice?

Grade 3

Code	What the student should be able to do	Example question
3.OA.A.1	Interpret products of whole numbers (multiplication as equal groups)	There are 4 bags with 6 apples each. How many apples total?
3.OA.A.2	Interpret whole-number quotients (division as sharing/grouping)	24 cookies are shared equally among 6 children. How many each?
3.OA.A.3	Multiply and divide within 100 to solve word problems	Each box holds 8 crayons. How many crayons in 7 boxes?
3.OA.C.7	Fluently multiply and divide within 100	What is 7×8 ?
3.OA.D.8	Solve two-step word problems using all four operations	A baker made 48 muffins and sold 15. He packs the rest in boxes of 3.
3.NBT.A.1	Round whole numbers to the nearest 10 or 100	Round 374 to the nearest 10.
3.NBT.A.2	Fluently add and subtract within 1,000	What is $856 - 478$?
3.NF.A.1	Understand a fraction $1/b$ as one part of a whole with b equal parts	A pie is cut into 8 equal slices. What fraction is 3 slices?
3.NF.A.2	Represent fractions on a number line	What fraction is the point halfway between 0 and 1?
3.NF.A.3	Explain fraction equivalence; compare fractions with same numerator/denominator	Which fraction is greater: $3/4$ or $3/8$?
3.MD.A.1	Tell time to the nearest minute; solve elapsed time problems	A movie starts at 2:15 and ends at 3:50. How long is the movie?
3.MD.C.5	Recognize area as an attribute; understand square units	A rectangle is 4 units wide and 3 units tall. What is its area?
3.MD.D.8	Solve problems involving perimeter of polygons	A rectangle is 6 cm long and 4 cm wide. What is its perimeter?
3.G.A.1	Understand shapes share attributes (e.g., all quadrilaterals have 4 sides)	What do all quadrilaterals have in common?

Grade 4

Code	What the student should be able to do	Example question
4.OA.A.1	Interpret a multiplication equation as a comparison	35 is 5 times as many as what number?
4.OA.A.3	Solve multi-step word problems; interpret remainders	A school buys 250 pencils for 38 students. How many extras are left over?
4.OA.B.4	Find factor pairs; identify prime and composite numbers within 100	Is 37 prime or composite?
4.NBT.A.2	Read and write multi-digit whole numbers; expand in base-ten	What is the value of the digit 6 in 364,891?
4.NBT.A.3	Round multi-digit whole numbers to any place	Round 47,382 to the nearest thousand.
4.NBT.B.4	Fluently add and subtract multi-digit whole numbers	What is $56,281 + 34,769$?
4.NBT.B.5	Multiply up to 4-digit by 1-digit; multiply two 2-digit numbers	What is 47×23 ?
4.NBT.B.6	Find quotients and remainders with up to 4-digit dividends	What is 6,372 divided by 4?
4.NF.A.1	Explain why $a/b = (n \times a)/(n \times b)$; generate equivalent fractions	What fraction is equivalent to $2/3$ with a denominator of 12?
4.NF.A.2	Compare fractions with different numerators and denominators	Which is greater: $5/8$ or $3/5$?
4.NF.B.3	Add and subtract fractions and mixed numbers with like denominators	What is $2/5 + 4/5$?

Code	What the student should be able to do	Example question
4.NF.C.6	Use decimal notation for fractions with denominators 10 or 100	Write 7/10 as a decimal.
4.MD.A.3	Apply area and perimeter formulas for rectangles in real-world problems	A rectangle is 12 m long and 8 m wide. What is its area?
4.MD.C.6	Measure angles in whole-number degrees using a protractor	What type of angle is 135 degrees?
4.G.A.2	Classify shapes based on parallel/perpendicular lines and angles	Which shape always has two pairs of parallel sides?

Grade 5

Code	What the student should be able to do	Example question
5.OA.A.1	Evaluate expressions with parentheses, brackets, or braces	What is $(3 + 4) \times 2 - 5$?
5.NBT.A.1	Understand place value: a digit is 10x the value of digit to its right	The digit 4 in 3,400 is how many times the value of 4 in 340?
5.NBT.A.3	Read, write, and compare decimals to thousandths	Which is greater: 0.45 or 0.405?
5.NBT.A.4	Round decimals to any place	Round 3.7284 to the nearest hundredth.
5.NBT.B.5	Fluently multiply multi-digit whole numbers	What is 347×82 ?
5.NBT.B.7	Add, subtract, multiply, and divide decimals to hundredths	What is $12.4 + 7.85$?
5.NF.A.1	Add and subtract fractions with unlike denominators	What is $\frac{2}{3} + \frac{3}{4}$?
5.NF.B.3	Interpret a fraction as division of the numerator by the denominator	What does $\frac{3}{4}$ mean as a division problem?
5.NF.B.4	Multiply fractions and mixed numbers	What is $\frac{2}{3} \times \frac{3}{5}$?
5.NF.B.7	Divide unit fractions by whole numbers and whole numbers by unit fractions	What is $\frac{1}{3}$ divided by 4?
5.MD.A.1	Convert measurement units within the same system	How many centimetres are in 3.5 metres?
5.MD.C.3	Recognize volume as an attribute of solid figures	A box is 3 cm x 4 cm x 5 cm. What is its volume?
5.G.A.1	Use a coordinate plane; plot points in the first quadrant	Plot the point (3, 5) on a coordinate grid. Which quadrant is it in?
5.G.B.3	Understand that attributes of a category apply to all subcategories	All squares are rectangles. True or false?

7. Edge Cases & Worked Examples

APPROVE	Grade 2 hard — $345 + 478$ flagged as Grade Mismatch Standard 2.NBT.7 explicitly covers adding within 1,000. Three-digit addition with regrouping IS the Grade 2 hard content. The AI flagged because the numbers look large, but it didn't account for the hard difficulty. Approve.
FLAG	Grade 2 fraction type — correct_answer '1/2', type 'fraction' Fraction input (the X/Y entry UI) is for Grade 3+ NF standards. A Grade 2 question should not use this type. Flag as UI Mismatch + Grade Mismatch. Suggested fix: change type to 'multiple_choice' with word answers, or move to Grade 3.
FLAG	Type numeric — correct_answer 'A half' The numeric keypad only accepts digits and a decimal point. 'A half' cannot be typed. Flag as UI Mismatch. Suggested fix: change correct_answer to '0.5' and type to 'numeric', or rephrase question and use 'multiple_choice'.
FLAG	Type multiple_choice — correct_answer '12' but options are ['ten', 'twelve', 'fourteen', 'sixteen'] The correct_answer '12' is a numeral but the options use words. No option value equals '12'. Flag as UI Mismatch. Suggested fix: either change correct_answer to 'twelve', or change option values to '10', '12', '14', '16'.
APPROVE	Grade 4 hard — 'Is 97 prime or composite?' flagged as Grade Mismatch 4.OA.B.4 explicitly covers prime and composite numbers. 97 is a valid choice for hard difficulty (it requires checking divisibility). Approve.
FLAG	Question says 'Look at the shape shown below.' No visual asset visible on card The student cannot answer without seeing the shape. Flag as Missing Visual. Note in Suggested Fix that a visual asset is needed (or rephrase the question to be self-contained).
FLAG	Correct answer verified wrong — 'What is 7×8?' with correct_answer '54' $7 \times 8 = 56$, not 54. Flag as Wrong Answer. Suggested fix: change correct_answer to '56'.
APPROVE	Multiple choice — one distractor is obviously wrong (e.g. '999' when other options are '12', '14', '16') The AI may flag this as Weak Distractors. But consider: for Grade 1-2 students, '999' is still a plausible wrong answer if they miscount. Use your judgement — if most distractors are reasonable, approve. Only flag if ALL wrong options are clearly absurd.

APPROVE	Grade 3 word problem with two steps — flagged as Grade Mismatch (too hard) 3.OA.D.8 explicitly requires two-step word problems. If it's tagged as hard difficulty, this is expected. Approve.
FLAG	Type fraction — correct_answer '3/0' Division by zero is undefined and the fraction UI rejects a denominator of 0. Flag as UI Mismatch + Wrong Answer. The question itself needs to be reworked.

Questions? Contact the MathKix team. Thank you for helping make MathKix great for kids.